

TOPIC: JOINS AND SET OPERATIONS

PART 1: CROSS JOIN

STEP 1: CREATE TABLES

```
CREATE TABLE class (  
    id INT,  
    name VARCHAR(30)  
);
```

```
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```

STEP 2: INSERT DATA

```
INSERT INTO class VALUES  
(1,'abhi'),  
(2,'adam'),  
(4,'alex');
```

```
INSERT INTO class_info VALUES  
(1,'DELHI'),  
(2,'MUMBAI'),  
(3,'CHENNAI');
```

QUESTION 1 - CROSS JOIN

SCENARIO:

School administration wants every student to be paired with every available city for survey analysis.

QUERY:

```
SELECT *
FROM class
CROSS JOIN class_info;
```

OUTPUT:

```
id name id address
```

1 abhi 1 DELHI

2 adam 1 DELHI

4 alex 1 DELHI

1 abhi 2 MUMBAI

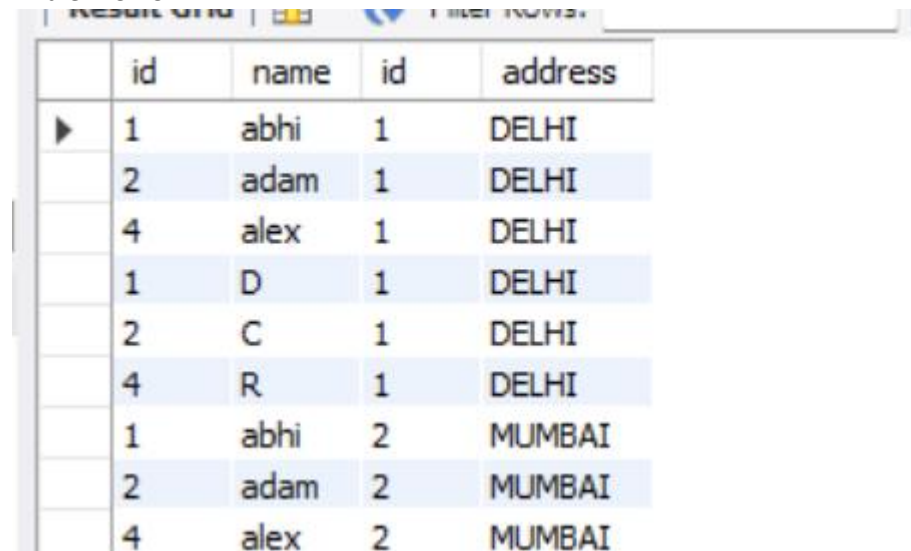
2 adam 2 MUMBAI

4 alex 2 MUMBAI

1 abhi 3 CHENNAI

2 adam 3 CHENNAI

4 alex 3 CHENNAI



	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	1	DELHI
	4	alex	1	DELHI
	1	D	1	DELHI
	2	C	1	DELHI
	4	R	1	DELHI
	1	abhi	2	MUMBAI
	2	adam	2	MUMBAI
	4	alex	2	MUMBAI

PART 2: INNER JOIN

```
CREATE TABLE class (
    id INT,
    name VARCHAR(30)
```

);

```
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```

```
INSERT INTO class VALUES  
(1,'abhi'),  
(2,'adam'),  
(3,'alex'),  
(4,'anu');
```

```
INSERT INTO class_info VALUES  
(1,'DELHI'),  
(2,'MUMBAI'),  
(3,'CHENNAI');
```

QUESTION 2 - INNER JOIN

SCENARIO:

Management wants only students whose address details are available in both tables.

QUERY:

```
SELECT *  
FROM class  
INNER JOIN class_info  
ON class.id = class_info.id;
```

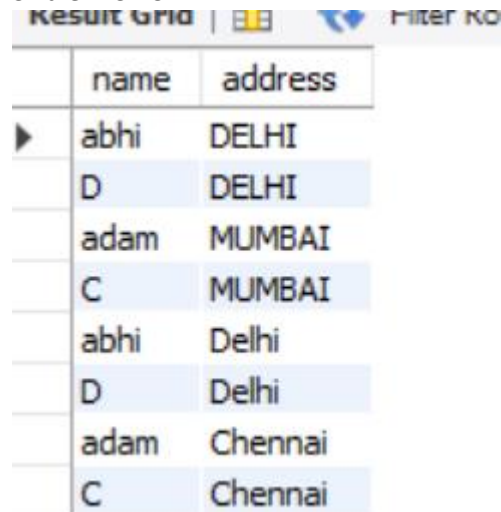
OUTPUT:

id name id address

1 abhi 1 DELHI

2 adam 2 MUMBAI

3 alex 3 CHENNAI



	name	address
▶	abhi	DELHI
	D	DELHI
	adam	MUMBAI
	C	MUMBAI
	abhi	Delhi
	D	Delhi
	adam	Chennai
	C	Chennai

QUESTION 3 - INNER JOIN

SCENARIO:

Find student names and addresses for registered students.

QUERY:

```
SELECT class.name,  
       class_info.address
```

```
FROM class
```

```
INNER JOIN class_info
```

```
ON class.id=class_info.id;
```

OUTPUT:

abhi DELHI

adam MUMBAI

alex CHENNAI

	id	name	address
	1	abhi	DELHI
	1	D	DELHI
	2	adam	MUMBAI
	2	C	MUMBAI
	1	abhi	Delhi
	1	D	Delhi
	2	adam	Chennai
	2	C	Chennai

PART 3: NATURAL JOIN

QUESTION 4 - NATURAL JOIN

SCENARIO:

The database administrator wants matching records automatically joined using common columns.

QUERY:

SELECT *

FROM class

NATURAL JOIN class_info;

OUTPUT:

id name address

1 abhi DELHI

2 adam MUMBAI

3 alex CHENNAI

	id	name	address
▶	1	abhi	DELHI
	1	D	DELHI
	2	adam	MUMBAI
	2	C	MUMBAI
	1	abhi	Delhi
	1	D	Delhi
	2	adam	Chennai
	2	C	Chennai

PART 4: LEFT OUTER JOIN

INSERT INTO class VALUES

(5,'ashish');

INSERT INTO class_info VALUES

(7,'NOIDA'),

(8,'PANIPAT');

QUESTION 5 - LEFT JOIN

SCENARIO:

School wants all students even if address information is not available.

QUERY:

SELECT *

FROM class

LEFT OUTER JOIN class_info

ON class.id=class_info.id;

OUTPUT:

1 abhi 1 DELHI

2 adam 2 MUMBAI

3 alex 3 CHENNAI

4 anu NULL NULL

5 ashish NULL NULL

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	Delhi
2	adam	2	MUMBAI
2	adam	2	Chennai
4	alex	NULL	NULL
1	D	1	DELHI
1	D	1	Delhi
2	C	2	MUMBAI
2	C	2	Chennai

QUESTION 6 - LEFT JOIN

SCENARIO:

Find students who do not have address records.

QUERY:

SELECT *

FROM class

LEFT JOIN class_info

ON class.id=class_info.id

WHERE class_info.id IS NULL;

OUTPUT:

4 anu

5 ashish

id	name	id	address
4	alex	NULL	NULL
4	R	NULL	NULL

PART 5: RIGHT OUTER JOIN

QUESTION 7 - RIGHT JOIN

SCENARIO:

Administration wants all address records even if students are missing.

QUERY:

SELECT *

FROM class

RIGHT OUTER JOIN class_info

ON class.id=class_info.id;

OUTPUT:

1 abhi 1 DELHI

2 adam 2 MUMBAI

3 alex 3 CHENNAI

NULL NULL 7 NOIDA

NULL NULL 8 PANIPAT

id	name	id	address
1	abhi	1	DELHI
1	D	1	DELHI
2	adam	2	MUMBAI
2	C	2	MUMBAI
NULL	NULL	3	CHENNAI
1	abhi	1	Delhi
1	D	1	Delhi
2	adam	2	Chennai
2	C	2	Chennai

QUESTION 8 - RIGHT JOIN

SCENARIO:

Find address records that do not belong to any student.

QUERY:

SELECT *

FROM class

RIGHT JOIN class_info

ON class.id=class_info.id

WHERE class.id IS NULL;

OUTPUT:

7 NOIDA

8 PANIPAT

	id	name	id	address
▶	NULL	NULL	3	CHENNAI
	NULL	NULL	3	Hyderabad

PART 6: FULL OUTER JOIN

QUESTION 9 - FULL OUTER JOIN

SCENARIO:

Management wants all records from both tables,
matched and unmatched.

QUERY:

SELECT *

FROM class

FULL OUTER JOIN class_info

ON class.id=class_info.id;

OUTPUT:

1 abhi 1 DELHI

2 adam 2 MUMBAI

3 alex 3 CHENNAI

4 anu NULL NULL

5 ashish NULL NULL

NULL NULL 7 NOIDA

NULL NULL 8 PANIPAT

Result Grid				Filter Rows:	
	id	name	id	address	
▶	1	abhi	1	DELHI	
	2	adam	2	MUMBAI	
	2	adam	2	Chennai	
	4	alex	NULL	NULL	
	1	D	1	DELHI	
	2	C	2	MUMBAI	
	2	C	2	Chennai	
	4	R	NULL	NULL	
	NULL	NULL	2	MUMBAI	